

Team Name: Alchemists

Track: AI Agents

Problem Statement: Build an AI-powered pull request review system that acts as an intelligent code reviewer. The platform helps repository maintainers by reading incoming pull requests, checking the code changes, tests, and structure, and giving one clear verdict on whether the pull request is safe to merge — through a set of autonomous AI agents.

Sub Part 1 (Planning - How it is being implemented): When a user submits a GitHub pull request link, the system plans out how to review it in steps. First, it fetches the pull request details and code changes. Then, it runs three specialized agents one after another — a Security agent, an Architecture agent, and a Testing agent — each checking a different part of the pull request. Once all three are done, their findings are combined to reach a final decision.

Sub Part 2 (Tool Usage - How it is being implemented): The agents use the GitHub API to gather real information about the pull request — the code changes, the description, and the test results GitHub already ran. The Security agent looks for risky code patterns. The Architecture agent checks if the changed code affects other files in the repository that weren't updated. The Testing agent reads the real test/CI results from GitHub. If the repository is private, the user logs in with GitHub so the system can access it; public repositories can be checked without logging in at all.

Sub Part 3 (Memory Management - How it is being implemented): The system remembers two kinds of information. Short-term memory holds everything gathered during one pull request review — the code changes, the test results, and each agent's findings — so they can be combined into a final answer. Long-term memory keeps a simple record of each repository's history, like files that have caused problems before, so the system gets better at spotting repeated risks in that repository over time.

Sub Part 4 (Autonomous Execution – How it is being implemented): Once the user submits a pull request link, the system runs the entire review on its own — no manual steps needed. Each agent checks its part, and once all three are finished, their findings are sent to an AI model whose only job is to write a clear, easy-to-read summary of the decision — Ready to Merge, Proceed with Caution, or Blocked. The AI does not make the decision itself; it only explains the decision the agents already made, so the result stays reliable and easy to trust.